

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287637

Kind Code

A1

Publication Date

September 11, 2025

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FLAT STI SURFACE FOR GATE OXIDE UNIFORMITY IN FIN FET DEVICES

Abstract

Operations in fabricating a Fin FET include providing a substrate having a fin structure, where an upper portion of the fin structure has a first fin surface profile. An isolation region is formed on the substrate and in contact with the fin structure. A portion of the isolation region is recessed by an etch process to form a recessed portion and to expose the upper portion of the fin structure, where the recessed portion has a first isolation surface profile. A thermal hydrogen treatment is applied to the fin structure and the recessed portion. A gate dielectric layer is formed with a substantially uniform thickness over the fin structure, where the recessed portion is adjusted from the first isolation surface profile to a second isolation surface profile and the fin structure is adjusted from the first fin surface profile to a second fin surface profile, by the thermal hydrogen treatment.

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Appl. No.: 19/214390

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 17322730 20210517 parent-grant-document US 12336217 child US 19214390

parent US continuation 16735495 20200106 parent-grant-document US 11011641 child US 17322730

parent US continuation 16230025 20181221 parent-grant-document US 10529863 child US 16735495

parent US continuation 15660355 20170726 parent-grant-document US 10192988 child US 16230025
parent US continuation 14925846 20151028 parent-grant-document US 9728646 child US 15660355
us-provisional-application US 62211682 20150828

Publication Classification

Int. Cl.: **H10D30/62** (20250101); **H10D30/01** (20250101); **H10D30/69** (20250101); **H10D62/10** (20250101); **H10D62/13** (20250101); **H10D62/822** (20250101); **H10D62/832** (20250101)

U.S. Cl.:

CPC **H10D30/6213** (20250101); **H10D30/024** (20250101); **H10D30/797** (20250101); **H10D62/116** (20250101); **H10D62/151** (20250101); **H10D62/822** (20250101); **H10D62/832** (20250101); **H10D62/8325** (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation patent application of U.S. patent application Ser. No. 17/322,730, filed on May 17, 2021, which is a continuation of U.S. patent application Ser. No. 16/735,495, filed on Jan. 6, 2020, now U.S. Pat. No. 11,011,641, which is a continuation of U.S. patent application Ser. No. 16/230,025, filed on Dec. 21, 2018, now U.S. Pat. No. 10,529,863, which is a continuation of U.S. patent application Ser. No. 15/660,355, filed on Jul. 26, 2017, now U.S. Pat. No. 10,192,988, which is a continuation of U.S. patent application Ser. No. 14/925,846, filed on Oct. 28, 2015, now U.S. Pat. No. 9,728,646, which claims the benefit of U.S. Provisional Patent Application No. 62/211,682, filed on Aug. 28, 2015, the subject matter of each of which is incorporated herein by reference in their entireties.

BACKGROUND

[0002] With increasing down-scaling of integrated circuits and increasingly demanding requirements to speed of integrated circuits, transistors need to have higher drive currents with increasingly smaller dimensions. Fin Field-Effect Transistors (Fin FET) were thus developed. Fin FETs include vertical semiconductor fins above a substrate. The semiconductor fins are used to form source and drain regions, and channel regions between the source and drain regions. Shallow Trench Isolation (STI) regions are formed to define the semiconductor fins. The Fin FETs also include gate stacks, which are formed with sidewalls and over the top surfaces of the semiconductor fins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the present disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be

limiting. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is an exemplary perspective view of a Fin Field-Effect Transistor (Fin FET) device in accordance with some embodiments of the present disclosure.

[0005] FIG. 2A and FIG. 2B are exemplary cross sectional views of the Fin FET device having a fin structure along a gate electrode in accordance with some embodiments of the present disclosure.

[0006] FIGS. 3, 4, 5, 6, 7, 8A, 8B, 9, 10, and 11 illustrate examples of cross-sectional views of intermediate stages of a sequential fabrication process of a Fin FET structure in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.”

[0009] FIG. 1 is an exemplary perspective view of a Fin Field-Effect Transistor (Fin FET) device **100** having a fin structure according to one embodiment of the present disclosure, and FIG. 2A is an exemplary cross sectional view of the Fin FET device **100** having a fin structure along a gate electrode according to one embodiment of the present disclosure. In these figures, some layers/features are omitted for simplification. FIG. 2B is a detailed view of a single fin in FIG. 2A.

[0010] The Fin FET device **100** depicted in FIGS. 1, 2A and 2B includes, among other features, a substrate **110**, a fin structure **120**, a gate dielectric layer **130** and a gate electrode layer **140**. In this embodiment, the substrate **110** is a silicon substrate. Alternatively, the substrate **110** may comprise another elementary semiconductor, such as germanium; a compound semiconductor including IV-IV compound semiconductors such as SiC and SiGe, III-V compound semiconductors such as GaAs, GaP, GaN, InP, InAs, InSb, GaAsP, AlGaN, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. In one embodiment, the substrate **110** is a silicon layer of an SOI (silicon-on insulator) substrate. When an SOI substrate is used, the fin structure **120** may protrude from the silicon layer of the SOI substrate or may protrude from the insulator layer of the SOI substrate. In the latter case, the silicon layer of the SOI substrate is used to form the fin structure **120**. Amorphous substrates, such as amorphous Si or amorphous SiC, or insulating material, such as silicon oxide may also be used as the substrate **110**. The substrate **110** may include various regions that have been suitably doped with impurities (e.g., p-type or n-type conductivity).

[0011] The fin structure **120** is disposed over the substrate **110**. The fin structure **120** may be made of the same material as the substrate **110** and may continuously extend from the substrate **110**. In

this embodiment, the fin structure **120** is made of silicon (Si). The silicon layer of the fin structure **120** may be intrinsic, or appropriately doped with an n-type impurity or a p-type impurity. [0012] In FIG. **1**, one fin structure **120** is disposed over the substrate **110**, while in FIG. **2A**, three fin structures **120** are disposed over the substrate **110**. However, the number of the fin structures is not limited to one or three. The numbers may be two or four or more. In addition, one or more dummy fin structures may be disposed in contact with both sides of the fin structures **120** to improve pattern fidelity in patterning processes. The width of the fin structure **120** is in a range of about 5 nm to about 40 nm in some embodiments, and may be in a range of about 7 nm to about 12 nm in certain embodiments. The height of the fin structure **120** is in a range of about 100 nm to about 100 nm in some embodiments, and may be in a range of about 50 nm to 100 nm in other embodiments.

[0013] In FIG. **2**, spaces between the fin structures **120** and/or a space between one fin structure and another element formed over the substrate **110** are filled by an isolation insulating layer (e.g., isolation region **150**) including one or more layers of insulating materials. The insulating materials for the isolation region **150** may include one or more layers of silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, fluoride-doped silicate glass (FSG), or a low-K dielectric material.

[0014] The lower part of the fin structure **120** under the gate electrode layer **140** is referred to as a well region **120A**, and the upper part of the fin structure **120** is referred to as a channel region **120B**, as shown in FIG. **2A**. Under the gate electrode layer **140**, the well region **120A** is embedded in the isolation region **150**, and the channel region **120B** protrudes from the isolation region **150**. A lower part of the channel region **120B** may also be embedded in the isolation region **150** to a depth of about 1 nm to about 5 nm. Although the upper corners of the fin structures **120** shown in FIG. **2** have right angles, this is only for an illustration purpose, and the shape of the upper corners of the fin structures **120** will be described later.

[0015] The channel region **120B** protruding from the isolation region **150** is covered by a gate dielectric layer **130**, and the gate dielectric layer **130** is further covered by a gate electrode layer **140**. Part of the channel region **120B** not covered by the gate electrode layer **140** functions as a source and/or drain of the Fin FET device **100** (see, FIG. **1**).

[0016] In certain embodiments, the gate dielectric layer **130** includes a single layer or alternatively a multi-layer structure, having one or more dielectric materials, such as a single layer of silicon oxide, silicon nitride, or high-k dielectric material, other suitable dielectric material, and/or combinations thereof, or a multilayer of two or more of these materials. Examples of high-k dielectric material include HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, other suitable high-k dielectric materials, and/or combinations thereof.

[0017] The gate electrode layer **140** includes one or more layers of any suitable material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof. The gate structure may be formed using a gate-last or replacement gate methodology.

[0018] In certain embodiments of the present disclosure, one or more work function adjustment layers **160** may be interposed between the gate dielectric layer **130** and the gate electrode layer **140**. The work function adjustment layer **160** may include a single layer or alternatively a multi-layer structure, such as various combinations of a metal layer with a selected work function to enhance the device performance (work function metal layer), liner layer, wetting layer, adhesion layer, metal alloy or metal silicide. The work function adjustment layers **160** are made of one or more conductive materials such as a single layer of Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, Re, Ir, Co, Ni, other suitable metal materials, or a multilayer of two or more of these materials. In some embodiments, the work function adjustment layer **160** may include a first metal material for the n-channel Fin FET and a second metal material for the p-

channel Fin FET. For example, the first metal material for the n-channel Fin FET may include metals having a work function substantially aligned with a work function of the substrate conduction band, or at least substantially aligned with a work function of the conduction band of the channel region **120B**. Similarly, for example, the second metal material for the p-channel Fin FET may include metals having a work function substantially aligned with a work function of the substrate valence band, or at least substantially aligned with a work function of the valence band of the channel region **120B**. In some embodiments, the work function adjustment layer **160** may alternatively include a polysilicon layer. The work function adjustment layer **160** may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. Further, the work function adjustment layer **160** may be formed separately for the n-channel Fin FET and the p-channel Fin FET which may use different metal layers.

[0019] Source and drain regions **125** are also formed in the upper part of the fin structure **120** not covered by the gate electrode layer **140**, by appropriately doping impurities in the source and drain regions **125**. An alloy of Si or Ge and a metal such as Co, Ni, W, Ti or Ta may be formed on the source and drain regions **125**.

[0020] In the formation of the isolation region **150**, etching operations are performed to recess a top surface **117** of the isolation region **150** to expose part of the fin structures **120**. As a result of the etching operation, in particular a wet etching process, a center portion of the top surface **117** between the fin structures **120** appears lower than the edges of the top surface **117**, which is adjacent to sidewalls **124** of the fin structures **120**, such that the top surface **117** has curved edges with a flat center. In this respect, the top surface **117** of the isolation region **150** is referred to as having a curved edge surface profile (see, FIG. 7).

[0021] The curved edge surface profile may inhibit formation of the gate dielectric layer **130** with a uniform thickness, which impacts the breakdown voltage of the Fin FET device **100** and induces an increase in leakage current across the gate dielectric layer **130**. In particular, the curved edge surface profile of the isolation region **150** may reduce the effective area of the channel region. In addition, the shape of the upper corners of the fin structures may affect the formation of gate oxide and the device performances. For example, the thickness of the gate dielectric layer **130** at top corners of the fin structure **120** becomes relatively smaller due to a Si orientation effect. Such thinning gate oxide may induce yield loss from plasma-induced damage.

[0022] The present disclosure provides a method of fabricating the Fin FET device **100** using a thermal hydrogen treatment operation after the formation of the isolation region **150** in the Fin FET device **100** such that the effects from the curved edge surface profile are decreased thereby improving performance of the Fin FET device **100** (see, FIG. 8). For example, physical dimensions of edge portions at the top surface **117** of the isolation region **150** can be decreased to minimize the curved edge surface profile. Such minimization of the curved edge surface profile increases an effective area of the channel region and facilitates gate oxide thickness conformity in the gate dielectric layer **130**. In addition, the thermal hydrogen treatment causes the top corners of the fin structure **120** to become substantially rounded (see, FIG. 8), which further facilitates the gate oxide thickness conformity thereby improving the breakdown voltage and minimizing the leakage current in the Fin FET device **100**.

[0023] FIGS. 3-11 illustrate examples of cross-sectional views of intermediate stages of a sequential fabrication process of a Fin FET device **100** in accordance with some embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after operations shown by FIGS. 3-11, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations may be interchangeable. The cross-sectional views in FIGS. 3-9 are taken under a gate, and the cross-sectional views in FIGS. 10 and 11 are taken at a source/drain region.

[0024] FIG. 3 is a cross-sectional view of the Fin FET device **100** at a first stage of various stages of fabrication according to an embodiment of the present disclosure. In this embodiment, the Fin

FET device **100** includes a substrate **110**. In one embodiment, the substrate **110** includes a crystalline silicon substrate (e.g., wafer). A p-type substrate or n-type substrate may be used and the substrate **110** may include various doped regions, depending on design requirements. In some embodiments, the doped regions may be doped with p-type or n-type dopants. For example, the doped regions may be doped with p-type dopants, such as boron; n-type dopants, such as phosphorus or arsenic; and/or combinations thereof. The doped regions may be configured for an n-type Fin FET, or alternatively configured for a p-type Fin FET.

[0025] In some alternative embodiments, the substrate **110** may be made of some other suitable elemental semiconductor, such as diamond or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. Also alternatively, the substrate may include an epitaxial layer. For example, the substrate may have an epitaxial layer overlying a bulk semiconductor. Further, the substrate may be strained for performance enhancement. For example, the epitaxial layer may include a semiconductor material different from that of the bulk semiconductor, such as a layer of silicon germanium overlying bulk silicon or a layer of silicon overlying bulk silicon germanium. Such strained substrates may be formed by selective epitaxial growth (SEG). Furthermore, the substrate may include a SOI structure. Also alternatively, the substrate may include a buried dielectric layer, such as a buried oxide (BOX) layer, such as that formed by separation by implantation of oxygen (SIMOX) technology, wafer bonding, SEG, or other appropriate process.

[0026] In one embodiment, a pad layer **304a** and a mask layer **304b** are formed on the semiconductor substrate **110**. The pad layer **304a** may be a thin film having silicon oxide formed, for example, using a thermal oxidation process. The pad layer **304a** may act as an adhesion layer between the semiconductor substrate **110** and the mask layer **304b**. The pad layer **304a** may also act as an etch stop layer for etching the mask layer **304b**. In at least one embodiment, the mask layer **304b** is formed of silicon nitride, for example, using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD). The mask layer **304b** is used as a hard mask during subsequent patterning processes. A photoresist layer **306** is formed on the mask layer **304b** and is then patterned by a photolithography patterning process. The photoresist layer may be removed after patterning of the mask layer **304b** and pad layer **304a** and before the trench etching.

[0027] FIG. **4** is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. The mask layer **304b** and pad layer **304a** are etched to expose underlying semiconductor substrate **110**. The exposed semiconductor substrate **110** is then trench-etched to form trenches **402** by using the patterned mask layer **304b** and pad layer **304a** as a mask.

[0028] In the trench etching process, the substrate **110** may be etched by various methods, including a dry etch, a wet etch, or a combination of dry etch and wet etch. The dry etching process may implement fluorine-containing gas (e.g., CF₄, SF₆, CH₂F₂, CHF₃, and/or C₄F₈), chlorine-containing gas (e.g., Cl₂, CHCl₃, CCl₄, and/or BCl₃), bromine-containing gas (e.g., HBr and/or CHBr₃), oxygen-containing gas, iodine-containing gas, other suitable gases and/or plasmas, or combinations thereof.

[0029] Next, a wet etching operation may be performed to remove a native oxide of the semiconductor substrate **110**. The wet etching operation may be performed using dilute hydrofluoric (DHF) acid. Portions of the semiconductor substrate **110** between trenches **402** form semiconductor fins **120**. The fins **120** may be arranged in strips (viewed from in the top of the Fin FET device **100**) parallel to each other, and closely spaced with respect to each other. Each of the fins **120** has a width W and a height H, and are spaced apart from an adjacent fin by a width S of the trench **402**. For example, the width W of the semiconductor fin **120** may be in a range of about 3 nm to about 30 nm in some embodiments.

[0030] FIG. 5 is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. Trenches **402** are filled with a dielectric material **502**. The dielectric material **502** may include a single layer or a multilayered structure having one or more dielectric materials, such as a single layer of silicon oxide, silicon nitride, or high-k dielectric material, other suitable dielectric material, and/or combinations thereof, or a multilayer of two or more of these materials. In one or more implementations, the dielectric material **502** is made of, for example, silicon dioxide formed by LPCVD (low pressure chemical vapor deposition), plasma-CVD or flowable CVD. In the flowable CVD, flowable dielectric materials instead of silicon oxide are deposited. Flowable dielectric materials, as their name suggests, can “flow” during deposition to fill gaps or spaces with a high aspect ratio. Usually, various chemistries are added to silicon-containing precursors to allow the deposited film to flow. In some embodiments, nitrogen hydride bonds are added. Examples of flowable dielectric precursors, particularly flowable silicon oxide precursors, include a silicate, a siloxane, a methyl silsesquioxane (MSQ), a hydrogen silsesquioxane (HSQ), an MSQ/HSQ, a perhydrosilazane (TCPS), a perhydro-polysilazane (PSZ), a tetraethyl orthosilicate (TEOS), or a silyl-amine, such as trisilylamine (TSA). These flowable silicon oxide materials are formed in a multiple-operation process. After the flowable film is deposited, it is cured and then annealed to remove un-desired element(s) to form silicon oxide. When the un-desired element(s) is removed, the flowable film densifies and shrinks. In some embodiments, multiple anneal processes are conducted. The flowable film is cured and annealed.

[0031] In some embodiments, other dielectric materials, such as silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or a low-K dielectric material, may also be used to form the dielectric material **502**. In an embodiment, the dielectric material **502** is formed using a high-density-plasma (HDP) CVD process, using silane (SiH_4) and oxygen (O_2) as reacting precursors. In other embodiments, the dielectric material **502** may be formed using a sub-atmospheric CVD (SACVD) process or high aspect-ratio process (HARP), in which process gases may include tetraethylorthosilicate (TEOS) and/or ozone (**03**). In yet other embodiments, the dielectric material **502** may be formed using a spin-on-dielectric (SOD) process, such as hydrogen silsesquioxane (HSQ) or methyl silsesquioxane (MSQ). In some embodiments, the filled recess region (or the trenches **402**) may have a multi-layer structure such as a thermal oxide liner layer filled with silicon nitride or silicon oxide.

[0032] FIG. 6 is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. After the deposition of the dielectric material **502**, planarization operations such as a chemical mechanical polish (CMP) and/or an etch-back process are then performed, followed by the removal of the mask layer **304b** and pad layer **304a**. An annealing process may be performed after the planarization operations. The annealing process includes rapid thermal annealing (RTA), laser annealing processes, or other suitable annealing processes.

[0033] In at least one embodiment, the mask layer **304b** is formed of silicon nitride such that the mask layer **304b** may be removed using a wet process using H_3PO_4 . The pad layer **304a** may be removed using dilute HF acid, if the pad layer **304a** is formed of silicon oxide. The remaining portions of the dielectric material **502** in the trenches **402** are hereinafter referred to as isolation regions **150**. In some embodiments, the removal of the mask layer **304b** and the pad layer **304a** is performed after the recessing of the isolation regions **150**, which recessing operation is shown in FIG. 7.

[0034] FIG. 7 is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. An etching process may be performed to etch isolation regions **150** to expose upper portions **722** of the semiconductor fins **120** from the isolation regions **150**. The etching process may include a dry etching process, wet etching process, or combination dry and wet etching processes to remove

portions of the isolation regions **150**. It is understood that the etching process may be performed as one etching process or multiple etching processes.

[0035] The remaining isolation regions **150** include top surfaces **717**. Further, the upper portions **722** of the semiconductor fins **120** protruding over the top surfaces **717** of the remaining isolation regions **150** thus are used to form part of an active area, such as a channel region, of the Fin FET device **100**. The upper portions **722** of the semiconductor fins **120** may include top surfaces **723** and sidewalls **724**. Height H1 of the upper portions **722** of the semiconductor fins **120**, which is to be a channel region, from the top surface **717** of the isolation regions **150** may be in a range of about 6 nm to about 100 nm. In some embodiments, the height H1 is greater than 100 nm or smaller than 6 nm.

[0036] As shown in FIG. 7, curved edge surface isolation structures **718** may be present at the top surfaces **717** of the isolation regions **150** sandwiched by two fins **120**. In the present disclosure, the curved edge surface isolation structures are defined by curved edges abutting the sidewalls **724** of the channel region **120B** (FIG. 1) with a flat center on the top surface **717**. The height ΔH of the curved edge surface isolation structures **718** can be in a range of $0.3H1 < \Delta H < 0.5H1$ in some embodiments, and in a range of $0.5H1 < \Delta H < 0.7H1$ in other embodiments.

[0037] The curved edge surface isolation structures **718** reduces surface area available in the channel region **120B** and causes non-uniform gate dielectric layer at the bottom of the channel region. The decrease in active surface area caused by the curved edge surface isolation structures **718** adversely impacts the collection of channel current flow. The non-uniform oxide thickness caused by the thinning effects also impacts the breakdown voltage of the Fin FET device **100** and induces an increase in leakage current across the gate oxide. The leakage current can be observed near the edges of the top surface **723** of the fins **120**, where the thickness of the gate oxide is smaller than that of the sidewalls **724** and/or the top surface **723**. In many cases, the variation in thickness across the surface of the channel region **120B** can vary by more than 0.6 nm. The thickness of the gate oxide at the edges of the top surfaces **723** of the semiconductor fins **120** becomes relatively smaller due to a Si orientation effect. Such thinning gate oxide may induce yield loss from plasma-induced damage.

[0038] In one or more implementations of the present application, a thermal hydrogen treatment is applied to the fin structure and the recessed oxide portion to improve the shape of the fin structure and to substantially eliminate the curved edge surface isolation structures **718** at the interface of the fin structure and the recessed oxide portion.

[0039] In this embodiment, the thermal hydrogen treatment is applied with hydrogen content ambient such as a hydrogen gas (H_2). The thermal hydrogen treatment may be applied with a gas flow rate in a range of about 0.2 standard-liter-per-minute (slm) to about 50 slm in some embodiments. The thermal hydrogen treatment may be applied with a temperature in a range of about 750 degrees Celsius to about 1000 degrees Celsius in some embodiments, and with a temperature in a range of about 850 degrees Celsius to about 1100 degrees Celsius in other embodiments. The thermal hydrogen treatment may be applied with a pressure in a range of about 0.2 Torr to about 760 Torr in some embodiments. The thermal hydrogen treatment may be applied for a duration in a range of about 20 seconds to about 120 seconds in some embodiments. In one or more implementations, the thermal hydrogen treatment is applied for a duration of 80 seconds.

[0040] FIGS. 8A and 8B are cross-sectional views of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. After the hydrogen treatment, the curved edge surface isolation structures can be diminished or eliminated as shown in FIGS. 8A and 8B and the upper corners of the fin structure can be rounded. For example, the semiconductor fins **120** are adjusted from a first fin surface profile (e.g., square upper edges) to a second fin surface profile (e.g., substantially round upper edges) based on the thermal hydrogen treatment. As shown in FIGS. 8A and 8B, the radius of the curvature R at the corners of the semiconductor fin **120** can be $0.05 W < R < 0.5 W$, W is a width of the channel region

of the semiconductor fin **120**, or $0.1\text{ W} < R < 0.2\text{ W}$. The change in height of the curved edge surface isolation structures **718** can be in a range of $0.05H_1 < \Delta H' < 0.1H_1$ in some embodiments, and in a range of $0.1H_1 < \Delta H' < 0.3H_1$ in other embodiments, where H_1 is the height of the upper portions **722** of the semiconductor fins **120**. In contrast to the upper edges of the fin structure prior to the thermal hydrogen treatment (see, FIG. 7), the fraction of the width and the fraction of the height that is curved after the thermal hydrogen treatment is significantly larger. Conversely, the bottom edges at the interface of the fin structure and recessed oxide portion (see, FIG. 7), the fraction of the height that is curved after the thermal hydrogen treatment is significantly smaller.

[0041] FIG. 9 is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the subject technology. A gate stack **920** is formed over the substrate **110** over the top surface **723** and sidewalls **724** of the semiconductor fin **120**, and extending to the top surfaces **717** of the first isolation region **150a** and the second isolation region **150b**. The gate stack **920** includes a gate dielectric layer **130** and a gate electrode layer **140** disposed on the gate dielectric layer **130**. During the formation of the gate stack **920**, various cleaning/etching processes, which etch the isolation regions **150**, may be performed. [0042] The gate dielectric **130** is formed to cover the top surface **723** and sidewalls **724** of at least a portion of the channel region of the semiconductor fins **120**. The gate dielectric formed on the upper surface of the isolation region is not shown in FIG. 9 for simplification. Since the upper corners of the fin structures are rounded by the thermal hydrogen treatment, the gate dielectric layer can have a thickness that is substantially uniform over the fin structure (no or less thinning effect). The uniformity of the gate dielectric layer **130** is also facilitated by the reduced curved edge surface profile of the isolation regions **150** at the bottom of the channel region (e.g., removed portions of the curved edge surface isolation structures **718**).

[0043] The top surface **717** of the isolation region may have a substantially flat step height as a result of the thermal hydrogen treatment. In addition, the top surface **723** of the semiconductor fin **120** may have substantially rounded corners as a result of the applied thermal hydrogen treatment. In one or more implementations, the substantially rounded corners are defined by a first curvature relating to a fraction of the width of the semiconductor fin **120** which is curved along the top surface **717** and a second curvature relating to a fraction of a height of the semiconductor fin **120** which is curved along the sidewalls **724**. For example, when the rounded corner is approximated by a part of a circle (in a cross-sectional view), the radius R of the circle (see, FIGS. 8A and 8B) is in a range of 0.1 W to 0.5 W in some embodiments, or 0.1 W to 0.2 W in other embodiments, where W is the width of the channel region **120B** in the semiconductor fin **120**. In contrast to the top surface **717** prior to the thermal hydrogen treatment (see, FIG. 7), the fraction of the width and the fraction of the height that is curved after the thermal hydrogen treatment is significantly smaller (see, FIGS. 8A and 8B). In this regard, the substantially flat STI surface enables to make the active surface area of the channel region larger and the substantially rounded corners of the semiconductor fin **120** enable to make the gate oxide uniform, thereby to improve the breakdown voltage and minimize the leakage current in the Fin FET device **100**.

[0044] In one or more implementations, the gate dielectric layer **130** has a thickness in a range of about 1.8 nanometers (nm) to about 5 nm, and a thickness in a range of about 2.0 nm to about 4.4 nm in some embodiments. The thickness of the gate dielectric layer **130** may be in a range of 3 nm to 5 nm in other embodiments. In one or more implementations, the gate dielectric layer **130** has a first thickness t_1 at the top surface **717** of the isolation region **150**, a second thickness t_2 at the top surface **723** of the semiconductor fin **120**, and a third thickness t_3 at one or more sidewalls **724** of the semiconductor fin **120**. If the gate dielectric layer **130** was formed on the structure shown in FIG. 7 without performing the thermal hydrogen treatment, the variation in thickness across the surface of the channel region **120B** would be more than 0.6 nm. In contrast, by adding the thermal hydrogen treatment, the variations of first thickness, the second thickness and the third thickness can be less than 0.3 nm. The variation of the thicknesses at the three identified locations may be

less than 0.6 nm in other embodiments. The uniform thickness of the gate dielectric layer **130** can minimize the yield loss from plasma induced damage.

[0045] In some embodiments, the gate dielectric layer **130** includes one or more layers of silicon oxide, silicon nitride, silicon oxy-nitride, or high-k dielectrics. High-k dielectrics may include metal oxides. Examples of metal oxides used for high-k dielectrics include oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Hf, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, and/or mixtures thereof. The gate dielectric layer **130** may be formed using a suitable process such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), thermal oxidation, UV-ozone oxidation, or combinations thereof. The gate dielectric layer **130** may include a single layer or multilayer structure. For example, the gate dielectric layer **130** may further include an interfacial layer (not shown) to reduce damage between the gate dielectric layer **130** and the fin **120**. The interfacial layer may include silicon oxide.

[0046] The gate electrode layer **140** is then formed on the gate dielectric layer **130**. In at least one embodiment, the gate electrode layer **140** covers the upper portion **722** of more than one semiconductor fin **120**, so that the resulting Fin FET device **100** includes more than one fin structure. In some alternative embodiments, each of the upper portions **722** of the semiconductor fins **120** may be used to form a separate Fin FET device **100**. The gate electrode layer **140** may include a single layer or a multilayer structure. The gate electrode layer **140** may include poly-silicon. Further, the gate electrode layer **140** may be doped poly-silicon with the uniform or non-uniform doping. In some alternative embodiments, the gate electrode layer **140** may include a metal such as Al, Cu, W, Ti, Ta, TiN, TiAl, TiAlN, TaN, NiSi, CoSi, other conductive materials with a work function compatible with the substrate material, or combinations thereof. The gate electrode layer **140** may be formed using a suitable process such as ALD, CVD, PVD, plating, or combinations thereof.

[0047] FIG. **10** is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. In this example, the cross-sectional view of the Fin FET device **100** is taken along an axis outside of the gate stack **920**.

[0048] The portion of the semiconductor fin **120** not covered by the gate electrode layer **140** is recessed to form a recessed portion **1026** of the semiconductor fin **120** having a top surface **1019** below the flat top surfaces **717** of the isolation regions **150**. In one embodiment, a biased etching process is performed to recess the top surface **1019** of the channel fin **722a** that are unprotected or exposed to form the recessed portion **1026** of the semiconductor fin **120**. In an embodiment, the etching process may be performed using HBr and/or Cl_{sub.2} as etch gases.

[0049] FIG. **11** is a cross-sectional view of the Fin FET device **100** at one of various stages of the sequential fabrication process according to an embodiment of the present disclosure. The structures depicted in FIG. **11** are produced by selectively growing a strained material **1130** over the recessed portion **1026** of the semiconductor fin **120** and extending over the top surfaces **717** of the isolation regions **150**. Since the lattice constant of the strained material **1130** is different from the channel region of the semiconductor fin **120**, the channel region of the semiconductor fin **120** is strained or stressed to enable carrier mobility of the device and enhance the device performance. Although the strained material **1130** are connected in FIG. **11**, the strained material **1130** may be formed separately with respect to each fin.

[0050] In at least one embodiment, the strained material **1130**, such as silicon carbon (SiC) and/or silicon phosphide (SiP), is epitaxially grown by a LPCVD process to form the source and drain regions of an n-type Fin FET. In at least another embodiment, the strained material **1130**, such as silicon germanium (SiGe), is epitaxially grown by a LPCVD process to form the source and drain regions of a p-type Fin FET. The p-type Fin FET and the n-type Fin FET are separately formed. In this regard, an n-type epitaxial region or p-type epitaxial region can be defined using photolithography and etching processes. For example, a protective layer, such as a silicon nitride

(SiN) layer, is formed to cover the p-type epitaxial region when the n-type epitaxial region is processed (e.g., S/D formation). Conversely, the SiN layer can cover the n-type epitaxial region when the p-type epitaxial region is processed.

[0051] The Fin FET device **100** serves only as one example. The Fin FET device **100** may be used in various applications such as digital circuit, imaging sensor devices, a hetero-semiconductor device, dynamic random access memory (DRAM) cell, a single electron transistor (SET), and/or other microelectronic devices (collectively referred to herein as microelectronic devices). Of course, aspects of the present disclosure are also applicable and/or readily adaptable to other type of transistor, including single-gate transistors, double-gate transistors, and other multiple-gate transistors, and may be employed in many different applications, including sensor cells, memory cells, logic cells, and others.

[0052] The present disclosure provides a method of fabricating the Fin FET device **100** using a thermal hydrogen treatment operation after the formation of the isolation region **150** in the Fin FET device **100** such that the effects from the curved edge surface profile are decreased thereby improving performance of the Fin FET device **100**. For example, physical dimensions of edge portions at the top surface of the isolation region **150** can be decreased to minimize the curved edge surface profile. Such minimization of the curved edge surface profile increases active surface area of the channel region and facilitates gate oxide thickness conformity at the bottom of the channel region. In addition, the thermal hydrogen treatment causes the top corners of the fin structure **120** to become substantially rounded, which further facilitates the gate oxide thickness conformity thereby improving the breakdown voltage and minimizing the leakage current in the Fin FET device **100**.

[0053] In an embodiment, a method of fabricating a Fin FET device is described herein. The method includes providing a substrate having a fin structure, an upper portion of the fin structure having a first fin surface profile. The method also includes forming an isolation region on the substrate and in contact with the fin structure. The method also includes recessing at least a portion of the isolation region by an etch process to form a recessed portion and to expose the upper portion of the fin structure, the recessed portion having a first isolation surface profile. The method also includes applying a thermal hydrogen treatment to the fin structure and the recessed portion. The method further includes forming a gate dielectric layer with a substantially uniform thickness over the fin structure, in which the recessed portion is adjusted from the first isolation surface profile to a second isolation surface profile and the fin structure is adjusted from the first fin surface profile to a second fin surface profile, by the thermal hydrogen treatment.

[0054] In another embodiment, a method of fabricating a Fin FET device is described herein. The method includes providing a substrate having fin structures, an upper portion of each of the fin structures having a first fin surface profile. The method also includes forming an isolation region on the substrate such that the fin structures are embedded in the isolation region. The method also includes recessing at least a portion of the isolation region by an etch process to form a recessed portion and to expose upper portions of the fin structures, the recessed portion having a first isolation surface profile between adjacent two fin structures. The method also includes applying a thermal hydrogen treatment to the fin structures and the recessed portion. The method further includes forming a gate dielectric layer with a substantially uniform thickness over the exposed upper portions of fin structures, in which the recessed portion is adjusted from the first isolation surface profile to a second isolation surface profile and the upper portions of the fin structures are adjusted from the first fin surface profile to a second fin surface profile, by the thermal hydrogen treatment.

[0055] In yet another embodiment, a semiconductor device includes a substrate and a fin structure having a top surface with substantially rounded corners, the substantially rounded corners being defined by a first radius of curvature relating to a fraction of a width of a channel region in the fin structure which is curved along the top surface and a second radius of curvature relating to a

fraction of a height of the channel region which is curved along sidewalls of the fin structure. The semiconductor device also includes an isolation region formed over the substrate and in contact with at least a sidewall of the fin structure, the isolation region having a top surface with a substantially flat step height based on a thermal hydrogen treatment, the substantially flat step height being defined by a downward slope from the sidewalls of the fin structure toward the top surface of the isolation region which relates to a fraction of the height of the channel region. The semiconductor device also includes a gate dielectric layer formed in contact with the fin structure and the top surface of the isolation region, the gate dielectric layer having a thickness that is substantially uniform over the fin structure and the isolation region based on the thermal hydrogen treatment.

[0056] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a fin disposed over a substrate, wherein an upper portion of the fin has a surface profile including a top surface of the fin having corners with a radius of curvature R , where $0.1W < R < 0.2W$, where W is a width of a channel region in the fin; an isolation insulating layer formed in contact with a sidewall of the fin, wherein the isolation insulating layer has a top surface having a step height ($\Delta H'$) being defined by $0.1H1 < \Delta H' < 0.3H1$, where $H1$ is a height of the channel region; and a gate dielectric layer has a thickness based on a thermal hydrogen treatment disposed over the top surface of the fin, the sidewall of the fin, and a top surface of the isolation insulating layer, wherein a thickness of the gate dielectric layer over the top surface of the fin and a thickness of the gate dielectric layer over the isolation insulating layer vary from each another by less than 0.3 nanometers.
2. The semiconductor device of claim 1, further comprising a gate electrode covering a portion of the fin.
3. The semiconductor device of claim 2, further comprising source and drain regions, each including a stressor layer disposed in and over recessed portions formed in the fin, the stressor layer applying a stress to a channel region of the fin covered by the gate electrode.
4. The semiconductor device of claim 3, further comprising an alloy of Si or Ge and Co, Ni, W, Ti, or Ta disposed over the source and drain regions.
5. The semiconductor device of claim 3, wherein the gate electrode comprises a material selected from the group consisting of polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, and combinations thereof.
6. The semiconductor device of claim 1, wherein the gate dielectric layer comprises a material selected from the group consisting of silicon oxide, silicon nitride, $\text{HfO}_{2.2}$, HfSiO , HfSiON , HfTaO , HfSiO , HfZrO , zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina ($\text{HfO}_{2.2}\text{—AlO}_{2.3}$) alloy, and combinations thereof.
7. The semiconductor device of claim 1, wherein the substrate comprises silicon, germanium, SiC, SiGe, GaAs, GaP, GaN, InP, InAs, InSb, GaAsP, AlGaN, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and combinations thereof.
8. A semiconductor device, comprising: a fin disposed over a substrate, an isolation insulating layer

formed in contact with a sidewall of the fin, wherein the isolation insulating layer has a top surface having a step height ($\Delta H'$) being defined by $0.1H_1 < \Delta H' < 0.3H_1$, where H_1 is a height of a channel region; and a gate dielectric layer has a thickness based on a thermal hydrogen treatment disposed over the top surface of the fin, the sidewall of the fin, and a top surface of the isolation insulating layer, wherein a thickness of the gate dielectric layer over the top surface of the fin and a thickness of the gate dielectric layer over the sidewall of the fin vary from each another by less than 0.3 nanometers.

9. The semiconductor device of claim 8, further comprising a gate electrode covering a portion of the fin.

10. The semiconductor device of claim 9, further comprising source and drain regions, each including a stressor layer disposed in and over recessed portions formed in the fin, the stressor layer applying a stress to a channel region of the fin covered by the gate electrode.

11. The semiconductor device of claim 10, further comprising an alloy of Si or Ge and Co, Ni, W, Ti, or Ta disposed over the source and drain regions.

12. The semiconductor device of claim 10, wherein the gate electrode comprises a material selected from the group consisting of polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, and combinations thereof.

13. The semiconductor device of claim 8, wherein the gate dielectric layer comprises a material selected from the group consisting of silicon oxide, silicon nitride, $\text{HfO}_{2.2}$, HfSiO , HfSiON , HfTaO , HfSiO , HfZrO , zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina ($\text{HfO}_{2.2}$ — $\text{AlO}_{2.3}$) alloy, and combinations thereof.

14. The semiconductor device of claim 8, wherein the substrate comprises silicon, germanium, SiC, SiGe, GaAs, GaP, GaN, InP, InAs, InSb, GaAsP, AlGaN, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and combinations thereof.

15. A semiconductor device, comprising: a fin, an isolation insulating layer formed in contact with a sidewall of the fin, wherein a channel region of the fin protrudes from the isolation insulating layer, the channel region of the fin has a surface profile including a top surface of the channel region having a radius of curvature R , where $0.05W < R < 0.5W$, where W is a width of the channel region, wherein the isolation insulating layer has a top surface having a step height ($\Delta H'$) being defined by $0.1H_1 < \Delta H' < 0.3H_1$, where H_1 is a height of the channel region; and a gate dielectric layer having a thickness based on a thermal hydrogen treatment disposed over the top surface of the fin, the sidewall of the fin, and a top surface of the isolation insulating layer, wherein a thickness of the gate dielectric layer over the top surface of the fin and a thickness of the gate dielectric layer over the sidewall of the fin vary from each another by less than 0.3 nanometers.

16. The semiconductor device of claim 15, further comprising a gate electrode covering a portion of the fin.

17. The semiconductor device of claim 16, further comprising source and drain regions, each including a stressor layer disposed in and over recessed portions formed in the fin, the stressor layer applying a stress to a channel region of the fin covered by the gate electrode.

18. The semiconductor device of claim 16, further comprising an alloy of Si or Ge and Co, Ni, W, Ti, or Ta disposed over the source and drain regions.

19. The semiconductor device of claim 16, wherein the gate electrode comprises a material selected from the group consisting of polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, and combinations thereof.

20. The semiconductor device of claim 15, wherein the gate dielectric layer comprises a material selected from the group consisting of silicon oxide, silicon nitride, $\text{HfO}_{2.2}$, HfSiO , HfSiON , HfTaO , HfSiO , HfZrO , zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina ($\text{HfO}_{2.2}$ — $\text{AlO}_{2.3}$) alloy, and combinations thereof.

